

Background

- Discrete Log
- Generic Group Model

Discrete logarithm

The dlog problem:

Given a generator g of a cyclic group G and a group element g^x .
Derive x

The CDH problem

Given two group elements g^x & g^y ,
compute g^{xy}

Generic Group Model

[Sho 97]

[Mau 05]

Generic group algorithms are algorithms that don't exploit any special structure of the of the group, & only uses abstract group operations.

Motivation

Let G be a cyclic group of order n generated by a generator g . G is isomorphic to the additive group $\mathbb{Z}_n$. A generic alg for computing the discrete log x of $b = g^x$, can be modeled as follows:

The algorithm is given a black-box which contains x . It can also input constants into the box and add values in the box. The only information reported back from the box is when an equality (collision) between two computed elements occurs. The algorithm's task is to extract x by provoking collisions and computing x from the collision pattern.

The Model:

B - a black box which can store values of a certain set S in internal state variables $V_1, \dots, V_m$.

B allows for two types of operations:

1. computation operations on internal state variables

For a set Π of operations on S , a computation operator consists of selecting an operation $f \in \Pi$ as well as indices $i_1, \dots, i_{t+1} \leq m$ of $t+1$ state variables.

B computes $f(V_{i_1}, \dots, V_{i_t})$ & stores the result in $V_{i_{t+1}}$.

2. Queries For a set Σ of relations on S , a query consists of selecting a relation $\phi \in \Sigma$ as well as the indices $i_1, \dots, i_t \leq m$ of t state vars. The query is replied by $\phi(V_{i_1}, \dots, V_{i_t})$.

3 types of problems:

Extraction : Extract the initial value x of V_1 .

Computation

Distinction

Simple Extraction problem in GGM

Theorem 1. Let $\star$ be a group operation on S , let $\Pi = C \cup \{x \mapsto x \star a \mid a \in S\}$ consist of all constant functions and multiplications by constants, and let $\Sigma = \{=\}$. The success probability of every k -step algorithm for extraction is upper bounded by $\frac{1}{4}k^2/|S|$, and by $km/|S|$ if m is bounded.

↑
max $|S|$

proof (sketch)

- As soon as a collision occurs between some state variables in the black box, the algorithm is successful.

$\forall x \in S$ and distinct a, b we have that
 $x \star a \neq x \star b$
group operation

Thus collision can occur only between operations of the form $x \mapsto x \star a$ and constant operations.

Let u and v be the corresponding number of operations the algorithm performs respectively.

Then probability of collision is upper bounded by $u \cdot v / |S|$.
Set $u = v = k/2 \Rightarrow$ gives first claim.

m finite: in each k steps # potential collisions is $m-1$.
Total number x for which this can occur is k .
 $\Rightarrow$ max $k \cdot (m-1)$ potential collisions.

Algebraic Group Model

Algebraic Group Model [FKL 19]

A model between standard model & GGM

Algebraic Algorithms

intuitively the only way for an alg algorithm to output a new group element is to derive it via group multiplication from known group elements

informally / intuitively

→ An algorithm A_{alg} is algebraic if whenever A_{alg} outputs a group element $z \in G$, it also outputs the "representation" $\vec{z} = (z_1, \dots, z_\ell) \in \mathbb{Z}_p^\ell$ such that $z = \prod d_i^{z_i}$, where $\vec{d} = (d_1, \dots, d_\ell)$ is the list of all group elements sent to A_{alg} .

Group elements are distinguished from other elements at a syntactical level. Which means that other elements must not depend on group elements.

Remark about AGM in RO model:

group elements output by the RO are included in the list $\vec{d}$.

Algebraic Group Model

- a computational model in which all adversaries are modeled as algebraic.

$H \Rightarrow_{\text{alg}} G$: Every algebraic adversary against G be transformed into an algebraic adv against H .
if H secure against alg adversaries so is G can adversary

Relation to GGM

AGM puts more restriction on adversary than the Standard Model
less GGM

To analyze the hardness of an assumption in the GGM, one must explicitly augment the model by any functionality that is offered by the structure of the group. As a simple example, let us consider a group G which is equipped with a symmetric bilinear map $e: G \times G \rightarrow GT$. The bilinear map can be modeled in the GGM via an oracle. However, it is not clear whether e can be used to gather even further information about the elements of G . Though it is widely believed that this is not the case, a proof in the GGM provides no answer to this question, because the GGM itself is based on the conjecture that e does not offer any functionality beyond a bilinear map.

No
similar
problems
in AGM

Let $G = (G, g, p)$
 cyclic group
 generator prime order

Lemma 2.2 Let G_G and H_G be algebraic security games such that $H_G \xrightarrow{(\Delta_\varepsilon, \Delta_t)}_{\text{alg}} G_G$ and winning H_G is (ε, t) -hard in the GGM. Then, G_G is $(\varepsilon \cdot \Delta_\varepsilon, t/\Delta_t)$ -hard in the GGM.

Computational Diffie Hellman

Theorem 3.1 $\text{dlog}_G \xrightarrow{(1,1)}_{\text{alg}} \{\text{cdh}_G, \text{sq-dh}_G\}$ and $\text{dlog}_G \xrightarrow{(3,1)}_{\text{alg}} \text{lc-dh}_G$.

Proof:

dlog_G^A

00 $x \xleftarrow{\$} \mathbb{Z}_p$
 01 $X := g^x$
 02 $z \xleftarrow{\$} A(X)$
 03 Return $(z = x)$

$\text{sq-dh}_G^{A_{\text{alg}}}$

00 $x \xleftarrow{\$} \mathbb{Z}_p$
 01 $X := g^x$
 02 $[Z]_{(a,b)} \xleftarrow{\$} A_{\text{alg}}(X)$
 03 Return $(Z = g^{x^2})$

Let A_{alg} be an algebraic adversary executed in the game sq-dh_G .

cdh_G^A

00 $x, y \xleftarrow{\$} \mathbb{Z}_p$
 01 $(X, Y) := (g^x, g^y)$
 02 $Z \xleftarrow{\$} A(X, Y)$
 03 Return $(Z = g^{xy})$

$\text{cdh}_G^{A_{\text{alg}}}$

00 $x, y \xleftarrow{\$} \mathbb{Z}_p$
 01 $(X, Y) := (g^x, g^y)$
 02 $[Z]_z \xleftarrow{\$} A_{\text{alg}}(X, Y)$
 03 Return $(Z = g^{xy})$

We construct a reduction R_{gen} as follows:

① on input X , run A_{alg} on X

② Upon receiving (Z, a, b) we have: $Z = g^a (g^x)^b = g^{x^2}$

Next we want to show reduction to cdh, thus we

show how to construct an adversary against cdh from an adversary against sq-dh.

③ solve for x in:

$$x^2 - bx - a = 0$$

④ check correct x (2 solutions) by comparing with X .

① Given $X = g^x$ sample $r \xleftarrow{\$} \mathbb{Z}_p$.

② Run the cdh adversary on (X, X^r)

③ Upon receiving Z , return $Z^{1/r}$.